Phase 1 Report

ProctorAI AI Based Online Proctoring System

This report defines the problem domain, reviews related work, checks project feasibility, and finalizes the datasets planned for the ProctorAI college project.

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Academic integrity note: The project description and planned features are based on our ProctorAI system work. Literature-based claims are supported through the references section. The wording and structure of this report were prepared with editorial assistance and should be reviewed by the team before submission.

1 Problem Domain and Problem Analysis

Online examinations are now used in colleges, training platforms, and recruitment processes. The main difficulty is that the candidate is not sitting in a controlled examination hall. In a physical exam, an invigilator can observe the candidate, check the surroundings, notice suspicious behavior, and take action immediately. In an online exam, the candidate may use another device, receive help from another person, leave the screen, copy content, switch tabs, or stop monitoring permissions without the examiner noticing at the right time.

The problem domain of this project is AI based online exam proctoring. The aim of ProctorAI is to monitor a candidate during an online test and generate evidence based integrity events when suspicious activity is detected. The system is not designed to replace the human proctor completely. Instead, it supports the proctor by collecting signals from the camera, microphone, screen recording, browser activity, and device checks.

The project is important because a normal online test page cannot protect exam integrity by itself. A practical system needs setup checks before the test, continuous monitoring during the test, and review tools after suspicious events are detected.

Risk Area	Example Problem	ProctorAI Response
Candidate identity and presence	Candidate leaves the camera frame or another person appears.	Detect face absent and multiple person events.
Unauthorized devices	A mobile phone or second screen is used during the test.	Detect phone objects and extra display conditions.
Browser misuse	Candidate copies, pastes, exits fullscreen, or switches away from the test.	Track browser integrity events with evidence.
Environment audio	Background voice or suspicious noise is present.	Monitor microphone signals and report audio events.
Evidence review	Automatic detection may be wrong in real conditions.	Send evidence to the proctor for human review.

2 Literature Survey and Existing Systems

Existing online proctoring systems usually follow one of three approaches: live proctoring, recorded proctoring, or automated AI based proctoring. Live proctoring depends on a human invigilator watching candidates in real time. Recorded proctoring stores video and activity logs for later review. Automated proctoring uses AI and browser signals to detect suspicious activity during the exam.

Research on AI based proctoring shows that common techniques include log tracking, face detection, object detection, and behavior analysis. These techniques are useful because cheating in an online exam can happen in different ways. Camera monitoring may detect face absence or a mobile phone, but it cannot detect copy paste activity by itself. Browser tracking can detect tab switching or shortcut keys, but it cannot detect another person in the room. This is why ProctorAI uses a multi signal approach.

Face detection and behavior analysis are frequently discussed in online proctoring studies. Object detection is also important because the system must recognize items such as mobile phones and the presence of more than one person. YOLO is suitable for this project because it is designed for real time object detection, which is needed while the candidate is actively taking the exam. The COCO dataset is also relevant because it is widely used for object detection and contains common object categories used in scene understanding.

Area Studied	What Existing Systems Use	Use in ProctorAI	Citation Needed
AI proctoring	Face detection, object detection, log tracking, behavior analysis.	Used as the overall system direction.	Yes
Face monitoring	Face detection and recognition for identity and presence checks.	Used for face absent and multiple face events.	Yes
Object detection	Real time detection models such as YOLO.	Used for phone and person detection.	Yes
Browser monitoring	Activity logs and event tracking.	Used for copy paste, focus lost, fullscreen exit, and shortcuts.	Yes
Human review	Proctor reviews suspicious events and evidence.	Used to reduce unfair automatic decisions.	Recommended

3 Proposed System Overview

ProctorAI is proposed as a web based AI online proctoring system for college examinations. It has two main parts: the candidate portal and the examiner or proctor portal. The candidate portal is used to select an assigned assessment, complete setup checks, and take the test. The proctor portal is used to create assessments, monitor candidate sessions, receive integrity events in real time, and review evidence.

The system will generate integrity events instead of directly declaring a candidate guilty. This is important because AI models can produce false positives due to lighting, camera position, background movement, or normal user behavior. A human proctor should be able to review the evidence before making a decision.

- Candidate portal: assignment selection, setup checks, screen recording setup, liveness verification, and test attempt.
- Proctor portal: assessment creation, session monitoring, real time integrity event feed, evidence playback, and event review.
- AI service: face presence analysis, multiple person detection, phone detection, and other camera based signals.
- Browser monitoring: copy paste, fullscreen exit, app focus lost, shortcut attempts, and screen recording status.

4 Feasibility Analysis

Technical feasibility: The project is technically feasible because modern browsers support camera, microphone, and screen recording permissions. A backend API can manage assessments, sessions, events, and evidence. Real time communication can be used to send integrity events to the proctor portal. AI models such as YOLO can support real time object detection.

Operational feasibility: The system fits college examination use because candidates can complete checks before starting the test and proctors can review suspicious events during or after the exam. This reduces the need for constant manual watching while still keeping human judgment in the process.

Economic feasibility: The project can be built using open source tools, browser APIs, and publicly available datasets. For a college project, the main cost is development time. If deployed publicly, storage for evidence clips and hosting would become the main operational costs.

Limitations: Browser based screen recording cannot silently capture the full screen without the user's permission. AI detection can also miss events or create false positives. Because of this, ProctorAI should support human review instead of making automatic punishment decisions.

5 Dataset Finalization

The following datasets and data sources are planned for ProctorAI. The goal is to use standard datasets for model support and a small custom validation set for realistic exam scenarios.

Dataset or Source	Purpose	Why It Is Suitable	Final Decision
COCO dataset	Object detection base data.	Contains labeled common objects and is widely used for object detection.	Use as base/reference dataset.
YOLO pretrained model data	Real time object detection.	Useful for detecting objects such as persons and phones during a live exam.	Use pretrained model and tune if needed.
Custom validation dataset	Project-specific testing.	Covers exam-like cases such as face absent, phone visible, multiple persons, poor lighting, and screen recording interruption.	Prepare a small validation set.
Browser event logs	Integrity event validation.	Captures copy paste, focus lost, fullscreen exit, and shortcut activity.	Generate during system testing.

6 Conclusion

The Phase 1 study confirms that AI based online proctoring is a relevant and feasible project domain. Online examinations need more than a basic test interface because candidates may use unauthorized devices, leave the camera, receive help, or misuse browser features. Existing research also shows that online proctoring systems commonly use face detection, object detection, log tracking, and behavior analysis.

ProctorAI will use a multi signal approach by combining camera monitoring, microphone monitoring, screen recording, browser activity tracking, connected display checks, liveness verification, and real time event reporting. The system will generate evidence based integrity events and allow the proctor to review them before making a final decision.

The finalized datasets and data sources are COCO, YOLO pretrained model data, a custom validation dataset, and browser event logs generated during testing. Based on the problem analysis, literature survey, and feasibility study, the project can continue into the design and implementation phase.

References

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4. Microsoft COCO Common Objects in Context. https://arxiv.org/abs/1405.0312